

MAHESH REDDY

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PROFESSIONAL SUMMARY

Machine Learning Engineer and AI Researcher (B.Tech, AI & Data Science, Class of 2027) with hands-on experience building and deploying ML systems across NLP, computer vision, and classification, achieving 92–99% model accuracy across projects. First-author published researcher on Emotionally Aware AI systems; GSSoC'26 open-source contributor with merged pull requests to production ML repositories. Proficient in Python, PyTorch, TensorFlow, and scikit-learn, with hands-on transformer and CNN architecture design experience.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science (Artificial Intelligence & Data Science)

Expected May 2027

Parul University, Gujarat, India | Dept. of AI and Data Science

Relevant Coursework: Machine Learning, Deep Learning, Artificial Neural Networks, Natural Language Processing (NLP), Computer Vision, Data Structures and Algorithms, OOP, DBMS

EXPERIENCE

GSSoC'26 Open Source Contributor May 2026 – Present

GirlScript Foundation | Open Source Program

- Shipped merged pull requests across 3+ production ML repositories, improving preprocessing modules, model training pipelines, and classification systems through code review and issue-tracking workflows.
- Earned GitHub Pull Shark recognition for sustained, high-quality open-source contribution.

ACHIEVEMENTS

- Nothing Playground Hackathon (Nothing)** – Selected to the Top 50 nationally for an original creation submitted to Nothing's Playground Hackathon.
- Vadodara Hackathon 6.0** – Placed in the Top 100 within Parul University's internal round.

PROJECTS

SentinelAI — Enterprise AI Governance Framework for Financial AI Systems 2026

Python, Pydantic, Local MLX LLM, Rule-Based Expert Systems, CLI (Typer/Rich) | [GitHub Repo](#)

- Engineered a governance CLI for financial AI systems that separates natural-language understanding from decision-making: a local on-device MLX model extracts a structured request, while every approve, review, or block decision is made by deterministic, versioned JSON rules.
- Designed six independent rule-driven expert domains (Policy, Fraud, Risk, Compliance, Spend, Audit) with full decision-rule traceability, plus chunked batch analysis for processing multiple requests via CLI or natural-language prompt.

MiniGPT-Emotional-Support V2 2026

Python, PyTorch, Transformer Architecture, Byte Pair Encoding (BPE) Tokenizer, Natural Language Processing (NLP) | [GitHub Repo](#)

- Implemented a GPT-style transformer from scratch (RMSNorm, SwiGLU, weight tying) to evaluate whether a small, locally trainable model could learn genuine empathetic response patterns.
- Trained the model on 52,137 pairs across three emotional-support corpora using a custom 16k-vocabulary BPE tokenizer, improving internal evaluation scores from 4/10 (V1) to 7.5/10 (V2).

Student Performance Risk Portal 2026

Python, scikit-learn, Pandas, REST API | [GitHub Repo](#) | [Live Demo](#)

- Deployed a scikit-learn classification pipeline behind a REST API to flag academic risk from habit and lifestyle data before grades slip, returning risk level, confidence scores, and actionable recommendations.
- Shipped the API live on Render for real-time use by students or advisors.

Sign Language Detection System 2026

Python, TensorFlow, Keras, Convolutional Neural Network (CNN), Computer Vision | [GitHub Repo](#)

- Engineered a baseline CNN and a transformer-based SignFormer model to translate hand gestures into letters in real time, lowering a communication barrier for hearing-impaired individuals.
- Integrated a MediaPipe hand-landmark pipeline for live webcam inference.

Customer Intelligence System 2026

Python, TensorFlow, Keras, scikit-learn, Pandas, FastAPI | [GitHub Repo](#)

- Engineered RFM behavioral features on UCI Online Retail data and evaluated K-Means clusters via silhouette scores to test whether combining classical clustering with a learned embedding space improves customer segmentation.
- Added a neural embedding layer exposed through a FastAPI backend for downstream use.

RESEARCH AND PUBLICATIONS

Emora — Local-First, Emotion-Aware AI Companion 2026

Research Project | Research Paper published on Zenodo

- Implemented a browser-based AI companion with a local-first inference pipeline using quantized Qwen3-4B on Apple Silicon, FastAPI, MongoDB, Qwen2-VL, Qwen3-TTS/Kokoro, and a WebGL VRM avatar.
- Designed selective persistent memory, bounded conversational context, deterministic emotion estimation, personality profiles, and a constrained Companion Brain for avatar behavior.
- Implemented privacy and safety controls including opt-in camera processing, user-controlled memory editing/export/deletion, bounded avatar actions, trusted-context separation, and non-diagnostic interaction boundaries.
- Verified the implementation with 170 automated tests: 169 passed and 1 skipped, alongside local runtime benchmarks for Qwen3-4B and TTS.
- Research paper: Zenodo — DOI: [10.5281/zenodo.22267180](https://doi.org/10.5281/zenodo.22267180)

Feature Scaling and Gradient Descent in Logistic Regression: A Breast Cancer Detection Case Study 2025

Independent Researcher, Self-Published

- Implemented the full gradient descent training pipeline from scratch in Python/NumPy and compared min-max normalization against standardization for convergence behavior on the Breast Cancer Wisconsin dataset (Kaggle), achieving 92%+ accuracy.

TECHNICAL SKILLS

Programming: Python, C, SQL

Machine Learning: scikit-learn, Feature Engineering, Hyperparameter Tuning, Model Evaluation, K-Means Clustering, Classification

Deep Learning: PyTorch, TensorFlow, Keras, NumPy, Pandas, Transformer Architecture, CNN, RNN, LSTM, Neural Networks

NLP & LLMs: Word2Vec, TF-IDF, BPE Tokenization, Text Classification, Sequence Modeling, Language Modeling, Generative Text Models

Computer Vision: Image Classification, Real-Time Object Detection, Image Preprocessing, Feature Extraction

Data Science: RFM Analysis, Behavioral Feature Engineering, Silhouette Score Evaluation, Data Preprocessing

Deployment & APIs: FastAPI, REST API Development, Render

Software Engineering: OOP, Data Structures and Algorithms, DBMS, Git, GitHub, Open-Source Development

Tools: Jupyter Notebook, VS Code, Google Colab, Kaggle

Research Areas: Multimodal AI, Emotion Recognition, Human-AI Interaction, Ethical AI Design, Curriculum Learning, Reinforcement Learning Concepts

CERTIFICATIONS AND TECHNICAL WRITING

CS50: Introduction to Programming with Python – Harvard University (edX) | 2025

Portfolio: maheshreddyml.netlify.app

AI/ML Technical Blog: throughmyeyesaiml.blogspot.com – in-depth articles on gradient descent, from-scratch ML, NLP architecture, and ethical AI design for software developer audiences.